



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

AIIMS Delhi Ransomware Attack(2022)

ITA1411-Ethical Hacking

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INTRODUCTION

What is Cyber Law?



Regulating Digital Activity

Cyber Law regulates activities involving computers, networks, and digital information.



IT Act, 2000

In India, the Information Technology Act, 2000 provides the legal framework for cybercrime.



Ethical Hacking

Helps organizations identify and fix vulnerabilities before attackers exploit them.



Healthcare Targets

Healthcare organizations are attractive targets because they store sensitive patient data.

ORGANIZATION OVERVIEW

About AIIMS Delhi

- AIIMS (All India Institute of Medical Sciences), New Delhi is one of India's leading public hospitals.
- Handles thousands of patients daily.
- Stores critical patient records and medical research data.
- Depends heavily on digital systems for healthcare operations.



PUBLIC HEALTHCARE

One of India's premier government-run medical institutions.



CRITICAL DIGITAL DATA

Patient records, research data, and hospital management systems.

THE CYBER ATTACK

AIIMS Delhi Ransomware Attack (2022)



**NOVEMBER
2022**



Hospital servers were infected with ransomware.



Patient registration and hospital management systems became unavailable.



Many services were temporarily shifted to manual operations.



The incident disrupted healthcare delivery and highlighted the importance of cybersecurity in critical infrastructure.

PREVENTIVE SECURITY MEASURES

Ethical Hacking Techniques That Could Help



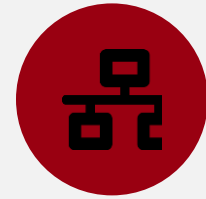
**Vulnerability
Assessment**



Penetration Testing



Security Auditing



Network Monitoring



Patch Management



Endpoint Protection



**Multi-Factor
Authentication (MFA)**



**Employee Security
Awareness Training**

LEGAL PROVISIONS

Relevant IT Act, 2000 Sections

Section 43	Unauthorized access and damage to computer systems.
Section 66	Computer-related offences.
Section 66C	Identity theft.
Section 66D	Cheating using computer resources.
Section 72	Breach of confidentiality and privacy.

CONSEQUENCES

Impact of the Attack



Disruption of hospital services.



Temporary loss of access to digital patient records.



Delays in patient care.



Increased operational costs.



Greater focus on cybersecurity in the healthcare sector.



Strengthened national cybersecurity discussions.

KEY TAKEAWAYS

Lessons Learned

- ✓ Conduct regular penetration testing.
- ✓ Keep systems updated with security patches.
- ✓ Back up critical data regularly.
- ✓ Implement Zero Trust security principles.
- ✓ Monitor networks continuously.
- ✓ Train employees to recognize phishing and other cyber threats.
- ✓ Prepare an incident response and disaster recovery plan.

BEST PRACTICES

Recommendations

-  Perform periodic vulnerability assessments.
-  Use strong access controls and MFA.
-  Deploy Endpoint Detection and Response (EDR).
-  Encrypt sensitive healthcare data.
-  Monitor network traffic using SIEM tools.
-  Follow CERT-In advisories and healthcare cybersecurity guidelines.
-  Conduct regular cybersecurity drills and audits.

Conclusion

- The AIIMS Delhi ransomware attack demonstrated the risks cyberattacks pose to critical healthcare infrastructure.
- Ethical hacking plays an essential role in discovering vulnerabilities before attackers exploit them.
- Compliance with the Information Technology Act, 2000, combined with proactive cybersecurity measures, helps reduce cyber risks.
- Continuous monitoring, regular testing, and employee awareness are key to protecting sensitive systems.